

PURBANCHAL UNIVERSITY

2024

Master of Computer Application (M.C.A.)/First Semester/Final

Time: 03:00 hrs.

Full Marks: 80 /Pass Marks: 32

MCA101: Discrete Structures (New Course)

MCA111: Discrete Mathematical Structures (Old Course)

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group A

Answer TWO questions.

2×16=32

1. Define binary operation. Explain the various properties of a binary operation. Prove $(\mathbb{Z}, +)$ is a group where $\mathbb{Z}$ is a set of integers and $+$ is an addition operation. 2+6+8
- 2(a) Let $A = \{1, 2, 3, 4\}$. The relation R on A is given as:
 $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 3), (1, 3), (4, 1), (4, 4)\}$
 - (i) Construct the diagram of R . 3
 - (ii) Obtain the matrix of relation R . 3
 - (iii) List the in degrees and out degrees of the relation R . 4
- (b) If a and b are positive integers such that $a=2296$ and $b=2952$. If $d=\text{GCD}(a, b)$, then express $d = sa + tb$ for some integers s and t . 6
- 3(a) Define euler circuit and path. Explain Fleury's algorithm to find euler circuit with an example. 4+12

Group B

Answer SEVEN questions.

7×8=56

4. How many committees of 5 people can be chosen from 20 men and 12 women if:
 - (i) Exactly 3 men must be on each committee?
 - (ii) At least 3 women must be on each committee?
5. Solve the recurrence relation $a_n = 7a_{n-1} - 10a_{n-2}$, $a_1=5$, $a_2=16$.

Contd. ...

6. Prove by using mathematical induction:

$$2+4+6+\dots+2n = n(n+1)$$

7. Let $P_1 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 3 & 2 & 1 & 4 & 5 & 6 \end{pmatrix}$ and $P_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 6 & 3 & 2 & 1 & 5 & 4 & 7 \end{pmatrix}$,

then compute: (i) $P_1 \circ P_2$ (ii) P^{-1}

(iii) Determine whether P_1 is even or odd permutation.

8. Explain group homomorphism and isomorphism with examples.
 9. Define Boolean Algebra. Explain POS and SOP of a boolean function. What is canonical form of a boolean function?
 10. Let $A = \{a, b, c, d, e\}$. A relation R on A is defined as:

$$R = \{(a, e), (b, a), (b, d), (c, d), (d, a), (d, c), (e, a), (e, b), (e, c), (e, e)\}.$$

Find transitive closure of R using Warshalls algorithm.

11. Determine the validity of the following argument by constructing the truth table.

If I study, then I will pass.

If I do not go to movie, then I will study.

I failed.

Therefore, I went to a movie.

12. Give the definition of minimal spanning tree. Use Kruskal's algorithm to find the minimal spanning tree for the given graph. List the edges in the order in which they are chosen.

